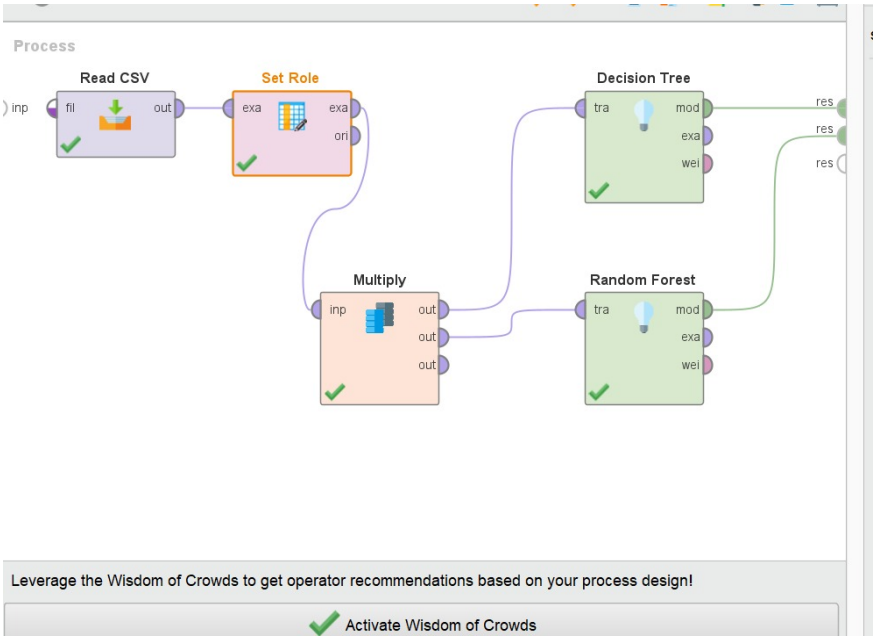
# Shape Prediction using Decision Tree and Random Forest in Altair AI Studio

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## Task

The data has **15 rows** with the columns **shape**, **color**, **outline** and **dot**, where the shape is known. The class work has two parts: **Part 1** builds a Decision Tree and a Random Forest from the 15 rows, and **Part 2** uses the Random Forest to predict the shape of 2 more rows whose shape is missing (?).

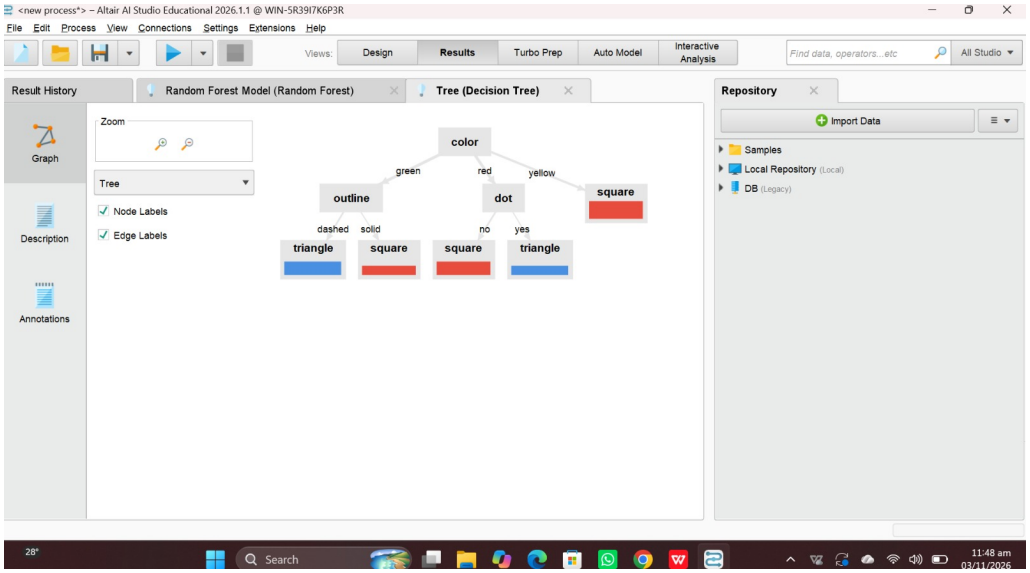
## Part 1: Decision Tree and Random Forest




Process design for Part 1.

1. **Read CSV:** loads the 15 rows.
2. **Set Role:** sets **shape** as the label, the column to predict. Color, outline and dot stay as regular inputs.
3. **Multiply:** makes two copies of the data, one for each model.
4. **Decision Tree:** builds a single tree from the data.
5. **Random Forest:** builds many trees from the same data and combines their results.

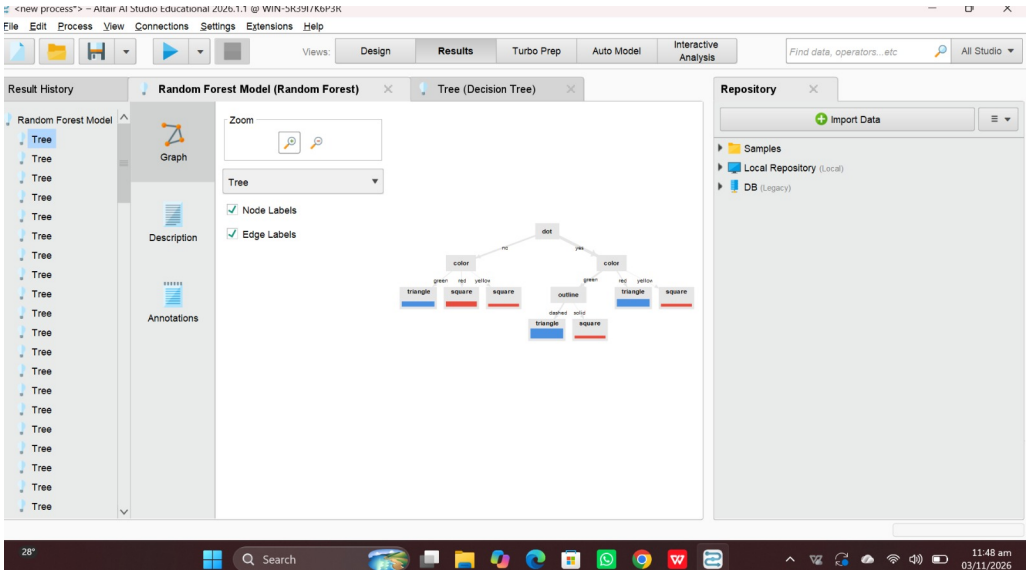
## Decision Tree Result




Decision Tree (Graph view).

The tree first asks the **color**. If it is **yellow**, the shape is a square. If it is **green**, it checks the **outline** (dashed = triangle, solid = square). If it is **red**, it checks the **dot** (no = square, yes = triangle).

## Random Forest Result

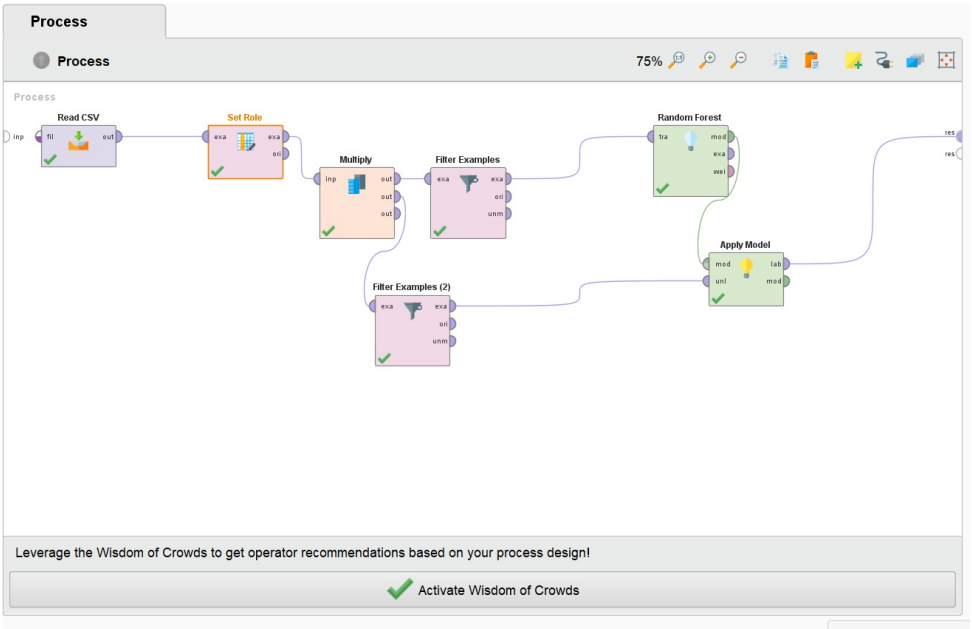



Random Forest model: the list on the left shows the individual trees, and the graph shows one selected tree.

The Random Forest builds many trees. Each tree uses a random part of the data, and the final answer is the combined vote of all trees. The tree shown starts with **dot** and then uses **color** and **outline**.

## Part 2: Predicting the Unknown Shapes

Two more rows have color, outline and dot, but their **shape is missing (?)**. The Random Forest trained on the 15 known rows is used to **predict the shape** of these 2 rows.




Process design for Part 2.

**1. Read CSV:** loads the data file.
**2. Set Role:** sets **shape** as the label. Color, outline and dot stay as regular inputs.
**3. Multiply:** makes two copies of the data, one for learning and one for prediction.
**4. Filter Examples (top):** keeps only the rows where the shape is known (the 15 rows) as training data.
**5. Random Forest:** learns from these rows by building many decision trees.
**6. Filter Examples (2):** keeps only the 2 rows where the shape is missing.
**7. Apply Model:** applies the trained Random Forest model (mod) to those 2 rows (unl) to predict their shape.

### Result

Open in Turbo Prep Auto Model Interactive Analysis

| Row No. | shape | prediction(s... | confidence(... | confidence(... | color | outline | dot |
|---|---|---|---|---|---|---|---|
| 1 | ? | square | 0.095 | 0.905 | green | solid | yes |
| 2 | ? | triangle | 0.898 | 0.102 | green | dashed | yes |

Apply Model output: predicted shape for the 2 unknown rows.

The shape column still shows **?** because it was unknown. The **prediction** column gives the answer, and the **confidence** columns show how sure the model is for each shape.

• Row 1 (green, solid, dot): predicted **square** with confidence 0.905.
• Row 2 (green, dashed, dot): predicted **triangle** with confidence 0.898.

**Conclusion:** using a Random Forest trained on the 15 known rows, the first unknown shape was predicted to be a **square** and the second a **triangle**.